

# MongoDB Queries

All queries used in the Student Management System project, run against the students collection.

## 1. Create Database

```
use Students
```

**Explanation:** Creates or switches to the Students database.

## 2. Create Collection

```
db.createCollection("students")
```

**Explanation:** Creates a collection named students.

## 3. Insert 30 Students

```
db.students.insertMany([ ... 30 documents ... ])
```

**Explanation:** Inserts all 30 student documents. Full data is in dataset.pdf.

## 4. Display All Students

```
db.students.find()
```

**Explanation:** Retrieves all documents from the students collection.

## 5. Equal To - \$eq

```
db.students.find({marks: {$eq: 88}})
```

**Explanation:** Finds students whose marks are exactly 88.

## 6. Greater Than - \$gt

```
db.students.find({marks: {$gt: 85}})
```

**Explanation:** Finds students whose marks are greater than 85.

## 7. Less Than - \$lt

```
db.students.find({marks: {$lt: 60}})
```

**Explanation:** Finds students whose marks are less than 60.

## 8. Greater Than or Equal To - \$gte

```
db.students.find({marks: {$gte: 90}})
```

**Explanation:** Finds students whose marks are 90 or above.

## 9. Less Than or Equal To - \$lte

```
db.students.find({marks: {$lte: 70}})
```

**Explanation:** Finds students whose marks are 70 or below.

## 10. Update Operation

```
db.students.updateOne(  
  {roll: 105},  
  {$set: {marks: 95}}  
)
```

**Explanation:** Updates the marks of the student with roll number 105 to 95.

## 11. Delete One

```
db.students.deleteOne({roll: 110})
```

**Explanation:** Deletes the document with roll number 110.

## 12. Delete Many

```
db.students.deleteMany({
  marks: {$lt: 60}
})
```

**Explanation:** Deletes all students having marks below 60 (rolls 111 and 113).

## 13. AND Operation - \$and

```
db.students.find({
  $and: [
    {marks: {$gt: 80}},
    {age: 19}
  ]
})
```

**Explanation:** Finds students satisfying both conditions.

## 14. OR Operation - \$or

```
db.students.find({
  $or: [
    {city: "Lucknow"},
    {city: "Delhi"}
  ]
})
```

**Explanation:** Finds students from Lucknow or Delhi.

## 15A. Sort Ascending

```
db.students.find().sort({marks: 1})
```

**Explanation:** Sorts students by marks from lowest to highest.

## 15B. Sort Descending

```
db.students.find().sort({marks: -1})
```

**Explanation:** Sorts students by marks from highest to lowest.

## 16. Limit Operation

```
db.students.find().limit(5)
```

**Explanation:** Displays only the first five documents.

## 17. Search by City

```
db.students.find({city: "Lucknow"})
```

**Explanation:** Finds all students whose city is Lucknow.

## 18. Count Documents

```
db.students.countDocuments()
```

**Explanation:** Counts documents currently in the collection (27 after the delete operations).

## MongoDB Operators Used

| Operator | Meaning                     | Example            |
|----------|-----------------------------|--------------------|
| \$eq     | Equal to                    | marks: {\$eq: 88}  |
| \$gt     | Greater than                | marks: {\$gt: 85}  |
| \$lt     | Less than                   | marks: {\$lt: 60}  |
| \$gte    | Greater than or equal to    | marks: {\$gte: 90} |
| \$lte    | Less than or equal to       | marks: {\$lte: 70} |
| \$and    | All conditions true         | \$and: [...]       |
| \$or     | At least one condition true | \$or: [...]        |